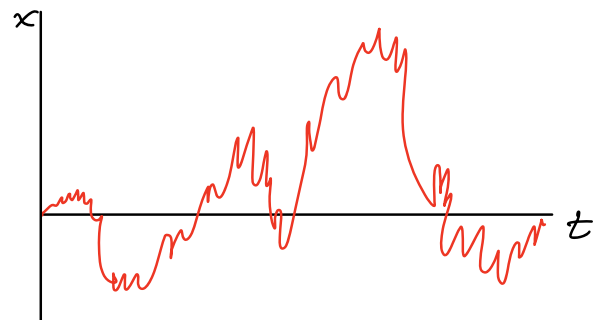


Recitation 2: The Wiener Process

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Wiener process $\{W(t) : t \geq 0\}$ is a process satisfying

- i) Continuous paths
- ii) Stationary, independent increments: $0 \leq t_1 < \dots < t_n$
 $[W(t_2) - W(t_1)], \dots, [W(t_n) - W(t_{n-1})]$ are independent
 $W(t_2) - W(t_1) \sim W(t_2 + s) - W(t_1 + s) \quad \forall s > 0.$
- iii) $W(t) \sim N(0, t) \quad \forall t \geq 0.$



b) Def Gaussian process $\{X(t)\}$

$\forall t_1, \dots, t_n \in \mathbb{R}^+$

Random vector $(X(t_1), \dots, X(t_n)) \in \mathbb{R}^n$ is multivariate Normal.

i.e. $\forall \{a_1, \dots, a_n\} \in \mathbb{R}, \sum_{i=1}^n a_i X(t_i)$ is Gaussian.

- e.g. $Y \sim N(0, 1)$
- $Z = -1$ with prob $1/2$ and 1 with prob $1/2$
- $W = ZY \sim N(0, 1)$
- vector (Y, W) is not multivariate normal ($Y + W$ is not Gaussian)

Wiener process is Gaussian:

show $\forall (a_1, \dots, a_n) \in \mathbb{R}$
 $(t_1, \dots, t_n) \in \mathbb{R}^+$

$\sum_{i=1}^n a_i W(t_i)$ is gaussian

e.g. let $t_1 < t_2$

$$\text{consider } W(t_1) + W(t_2) = (W(t_1) - W(0)) + W(t_2) - W(t_1) + W(t_1)$$

$$= 2(W(t_1) - W(0)) + (W(t_2) - W(t_1))$$

$\Rightarrow$ Gaussian. $\nwarrow$ indep, normal $\nearrow$

Any Gaussian process $X(t)$ is completely determined by its mean $\langle X(t) \rangle$

and covariance $\text{cov}(X(t_1), X(t_2)) = \langle X(t_1) X(t_2) \rangle_c \equiv \langle X(t_1) X(t_2) \rangle - \langle X(t_1) \rangle \langle X(t_2) \rangle$

let's compute for $t, s > 0$

$$\begin{aligned}
\langle W(t) W(t+s) \rangle &= \langle W(t) (W(t+s) - W(t) + W(t)) \rangle \\
&= \langle W(t)^2 \rangle + \langle W(t) (W(t+s) - W(t)) \rangle \\
&= t
\end{aligned}$$

For general times t_1, t_2 :

$$\langle W(t_1) W(t_2) \rangle = \min(t_1, t_2).$$

[Alternative definition for Wiener process: a Gaussian process with mean 0 and covariance $\langle W(t_1) W(t_2) \rangle_c = \min(t_1, t_2)$.

c) Langevin equation

$$\begin{aligned}
\dot{x} &= \eta(t), & \langle \eta(t) \eta(t') \rangle &= \delta(t-t') \\
x(t) &= \int_0^t ds \eta(s) \Rightarrow \text{check Gaussian process}
\end{aligned}$$

$$\begin{aligned}
\langle x(t) x(t+s) \rangle &= \int_0^t \int_0^{t+s} du_1 du_2 \frac{\langle \eta(u_1) \eta(u_2) \rangle}{\delta(u_1 - u_2)} = \int_0^t du_2 = t. \\
&\Rightarrow \text{Wiener process}
\end{aligned}$$

d) Let $W(t)$ be a Wiener process. The following are also Wiener processes:

$$X(t) \equiv c^{-1/2} W(ct) \rightarrow \text{"scale invariant", fractal.}$$

$$Y(t) \equiv W(c+t) - W(c)$$

$$Z(t) \equiv t W(1/t)$$

Check X : clearly Gaussian. $\langle X(t_1) X(t_2) \rangle = c^{-1} \min(ct_1, ct_2) = \min(t_1, t_2)$.

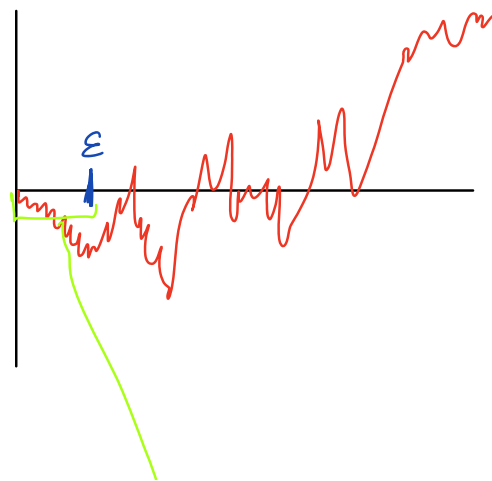
Check Z : Gaussian. $\langle Z(t_1) Z(t_2) \rangle = t_1 t_2 \min(1/t_1, 1/t_2) = \frac{t_1 t_2}{\max(t_1, t_2)} = \min(t_1, t_2)$.

Some pathologies of Brownian motion

1) $\forall \varepsilon > 0$, $W(t)$ crosses zero infinitely many times

Why? Consider $Z(t) = t W(1/t)$ on the interval $(1/\varepsilon, \infty)$.

$$Z(t) = 0 \Leftrightarrow W(1/t) = 0 \text{ for } t > 0.$$

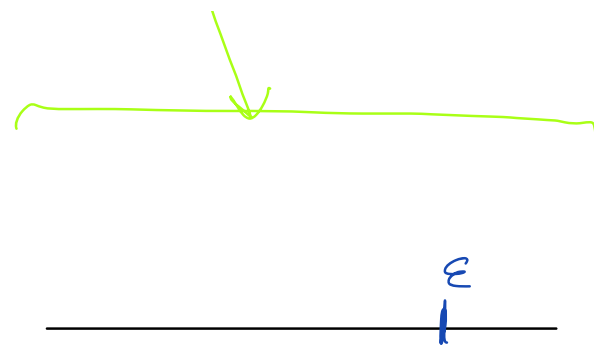


2) $W(t)$ is nowhere differentiable with prob 1.

Consider $\Delta W = W(t+\delta) - W(t)$

$$\langle \Delta W^2 \rangle = \delta$$

hence $\lim_{\delta \rightarrow 0} \frac{\Delta W}{\delta} \sim \frac{\sqrt{\delta}}{\delta} \rightarrow \infty$.



First passage time

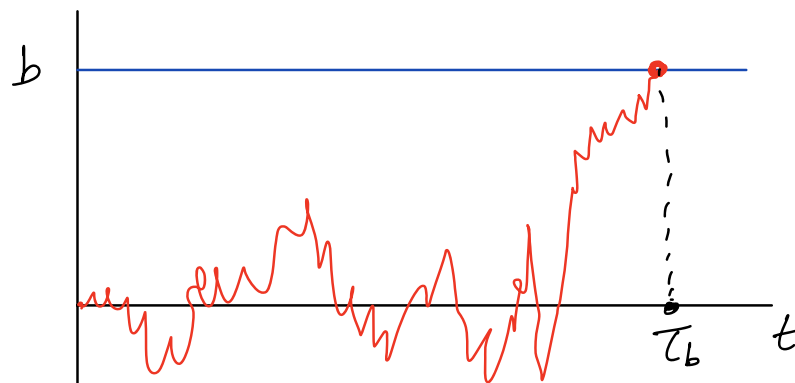
Let $\{W(t)\}$ be a Wiener process

Def

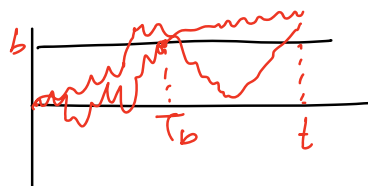
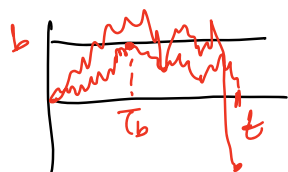
$$\tau_b \equiv \inf \{t : W(t) \geq b\}$$

$$= \inf \{t : W(t) = b\}$$

Compute cumulative dist: $F(t) \equiv \mathbb{P}[\tau_b \leq t] = \int_0^t ds P_{\tau_b}(s)$ ← prob density.

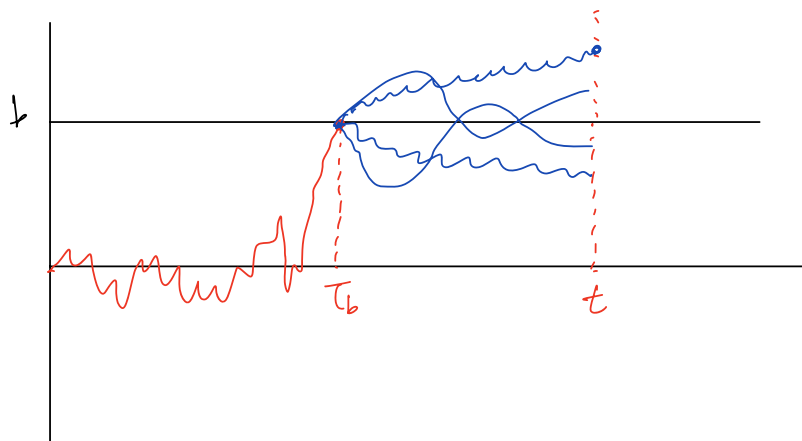


$$F(t) = \mathbb{P}[\tau_b \leq t, W(t) < b] + \mathbb{P}[\tau_b \leq t, W(t) > b]$$



$$\mathbb{P}[\tau_b \leq t, W(t) > b] = \mathbb{P}[W(t) > b] \quad (\text{Second term})$$

$$\mathbb{P}[\tau_b \leq t, W(t) < b] = \underbrace{\mathbb{P}[W(t) < b | \tau_b \leq t]}_{1/2} \underbrace{\mathbb{P}[\tau_b \leq t]}_{F(t)} \quad (\text{First term}).$$



$$F(t) = \frac{1}{2} F(t) + \underbrace{P[w(t) > b]}_{N(0, t)}$$

$$F(t) = 2 \int_b^{\infty} \frac{e^{-s^2/t}}{\sqrt{2\pi t}} ds \quad \int_b^{\infty} ds \frac{e^{-s^2/t}}{\sqrt{2\pi t}}$$

Probability density:

$$P_{T_b}(t) = F'(t) = \frac{\partial}{\partial t} \int_{b/\sqrt{t}}^{\infty} \frac{e^{-u^2}}{\sqrt{2\pi}} du = \frac{b e^{-b^2/2t}}{\sqrt{2\pi t^3}} \quad (\text{for } t > 0)$$

Normalizable, but $\langle T_b \rangle = \infty$

Levy dist



We say that a Brownian motion in 1D is "null-recurrent": if it crosses zero, it comes back almost surely, but the time to do so is infinite on average. (imagine $b \rightarrow 0$)

Most likely hitting time: $\arg\max p(t) = b^2/3$.

Consistent with diffusive scaling $\Delta x^2 \sim t$

General Brownian motions

We may generalize the Wiener process by adding a drift and a scale to $W(t)$.

A process $X(t)$ defined as

$$X(t) = X(0) + \mu t + \sigma W(t)$$

is called a (μ, σ^2) -Brownian motion (though this terminology is rarely used in physics).

I bring this up because of an important fact:

Thm: If a stochastic process X has continuous paths and stationary independent increments, then X is a Brownian motion as defined above.

The Gaussianity of increments thus comes "for free". This can be intuited from the central limit theorem: If increments are stationary and independent, then any one increment can be partitioned as the sum of many smaller increments that are i.i.d.